

Driving Behavior Analysis from Real OBD-II Telemetry

Technical Report | Setayesh Taherzadeh |
github.com/setayeshtaherzadeh/vehicle-driving-behavior-analysis

1. Motivation

This project investigates a concrete question in intelligent vehicle systems: can risky driving moments (harsh braking, harsh acceleration) be detected directly from raw, unlabeled vehicle sensor data, and can that detection be turned into something a driver could actually see and react to? It uses real OBD-II telemetry rather than a simulated dataset, and deliberately avoids machine learning in favor of a transparent, statistically grounded method — the aim was to get the fundamentals of detection and human-facing presentation right before adding complexity. This was built as part of preparation for graduate study in automotive software and human-machine interaction (HMI).

2. Dataset

The data is real-world OBD-II telemetry from a 2012 Opel Corsa 1.2 (A12XER, 84 hp), collected over roughly 100 hours of driving between April 2025 and June 2026 (source: *OBD2_panel_opel_2012*, Kaggle, CC BY 4.0). The raw file contains about 789,000 timestamped readings across 22 driving periods (each spanning several individual trips between refuels), with 31 recorded columns.

Cleaning kept the 11 columns relevant to this analysis (speed, RPM, throttle, coolant temperature, gear, fuel usage, and period identifiers); columns such as *TORQUE* and *POWER* had over 78% missing values from ECU polling gaps and were dropped. A small number of *SPEED* readings sitting exactly at 255 — the sensor's upper bound and a known invalid-reading placeholder — were removed. The result is 789,266 rows (99.99% of the original data) sorted chronologically within each period.

3. Method

Speed alone says little about risk; what matters is how quickly it changes. Instantaneous acceleration was computed for each reading as the change in speed divided by the change in time since the previous reading, calculated separately within each driving period so that the last reading of one period is never compared against the first reading of the next.

Rather than an arbitrary fixed cutoff, the harsh-event threshold was defined statistically: the bottom 1% and top 1% of the acceleration distribution were flagged as harsh braking and harsh acceleration, respectively (braking threshold < -6.85 , acceleration threshold > 4.47). This threshold is data-driven and reproducible rather than a subjective guess. In total, 15,786 harsh events were flagged (2.00% of all readings).

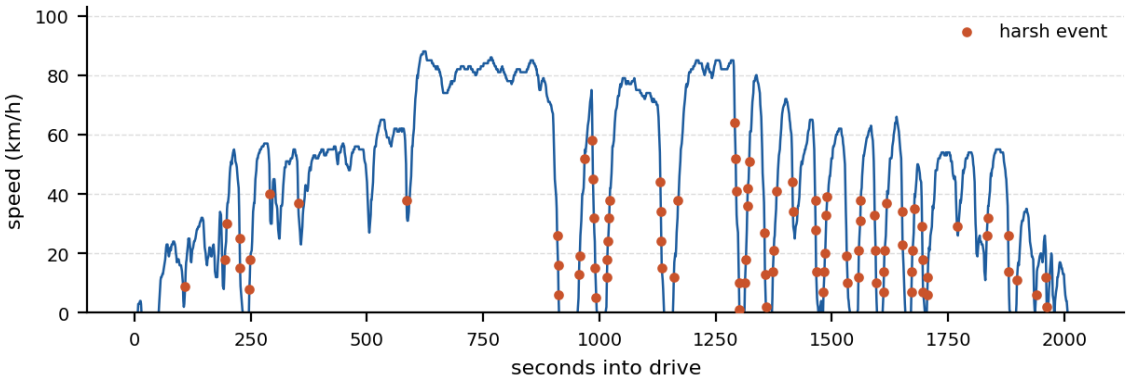


Figure 1. Speed trace of one recorded drive, with harsh events marked.

4. Key Finding

As a validation check, the harsh-event rate was compared against average fuel usage across the 22 periods. The result ran counter to the initial hypothesis: **correlation = -0.37** — harsher periods used *less* fuel per minute, not more.

The confound turned out to be average speed, which correlates strongly with fuel use ($+0.94$) but negatively with harsh-event rate (-0.29). Highway-style driving sustains higher engine load and burns more fuel per minute, but is smoother with fewer sudden inputs; city driving produces more harsh events from stop-and-go traffic, but lower fuel use per minute since the engine spends more time idling. Harsh-event rate alone is therefore not a reliable predictor of fuel consumption — driving context matters more than event count.

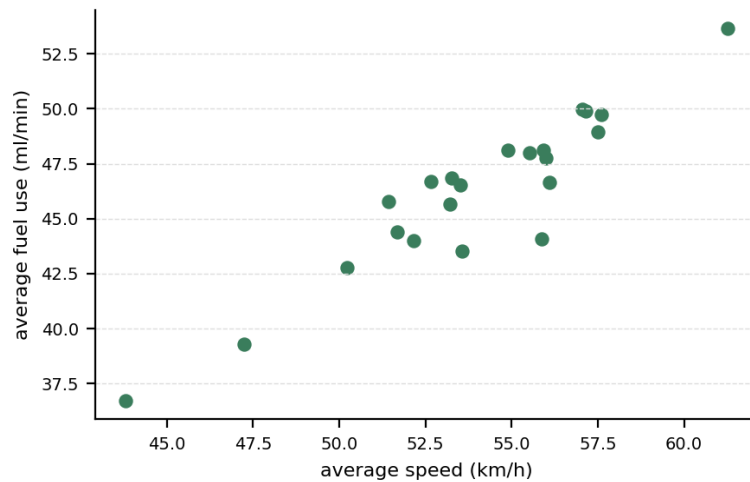


Figure 2. Average speed vs. average fuel use across the 22 driving periods.

5. From Detection to Interaction

The detection method above answers whether a risky moment can be identified from raw sensor data at all, but says nothing about how that information would reach a driver. To connect the two, the accompanying dashboard includes a replay simulation: it plays one recorded drive back second-by-second at adjustable speed and surfaces an on-screen warning the instant a harsh event is detected — a proof of concept for how this detection logic could feed a real-time in-car alert.

This is deliberately scoped as a proof of concept, not a production HMI: it runs as a browser-based web app, not on an in-vehicle display; it replays recorded data rather than a live sensor feed; the warning is a single visual style with no auditory channel; and no real driver has seen or reacted to it. These are named directly as next steps below rather than presented as solved.

6. Limitations & Future Work

- The 1% threshold is transparent and reproducible, but has no notion of context — a harsh brake in traffic is not the same as one on an empty road.
- The vehicle has no ADAS, so nothing here measures how a real driver responds to a warning, only whether one can be triggered.
- The current warning is visual-only and untested with real drivers; multi-modal alerts (visual + audio) are standard practice in safety-critical HMI and a natural next step.
- Natural extensions include a richer driver profile using RPM/throttle/gear behavior, and a small user study comparing warning designs (timing, modality, salience) against driver comprehension and reaction time.